VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

Student Name	Registration Number
Shaurya Vardhan	24BCE11397
Slot	**Date of Submission**
E22	

1. Introduction

In today's digital world, web browsers play a crucial role in accessing information, communication, and online services. Every activity on the internet, from searching for information to using web applications, depends on a browser. Among the many browsers available, Mozilla Firefox stands out as one of the most important open-source web browsers. Mozilla Firefox is developed by the Mozilla Foundation and is known for its strong emphasis on privacy, security, and openness. Unlike proprietary browsers, Firefox is built with the philosophy of keeping the web open and accessible to everyone. This report provides a detailed audit of Mozilla Firefox as an open-source software project. It explores its origin, licensing model, ethical considerations, Linux footprint, ecosystem, and comparison with proprietary alternatives.

2. Choosing Your Software

Pick one open-source project from the list below, or propose your own (subject to instructor approval). Your choice will be the subject of your entire report and all five shell scripts.

Approved software options

Software	Category	License	Why it's interesting
Linux Kernel	Operating System	`GPL v2`	*The foundation everything else runs on*
Apache HTTP Server	Web Server	`Apache 2.0`	*Powers ~30% of all websites globally*
MySQL	Database	`GPL v2 / Commercial`	*A license with a dual-model story to tell*
Firefox	Web Browser	`MPL 2.0`	*A nonprofit fighting for an open web*
VLC Media Player	Multimedia	`LGPL / GPL`	*Plays anything — built by students in Paris*
LibreOffice	Office Suite	`MPL 2.0`	*Born from a community fork — a real lesson*
Python	Programming Language	`PSF License`	*A language shaped entirely by community*
Git	Version Control	`GPL v2`	*The tool Linus built when proprietary failed him*

Your choice	Write your chosen software here before submission: Firefox

2. Project Report Structure

3.

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem Firefox Was Created To Solve

In the late 1990s and early 2000s, the internet browser market was dominated by Microsoft Internet Explorer. It came pre-installed with Windows, which gave Microsoft a massive advantage and led to near monopoly in the browser space. However, Internet Explorer had several limitations: Poor performance and frequent crashes Serious security vulnerabilities Lack of support for modern web standards Limited customization for users Strong control by a single corporation This situation raised concerns among developers and the open-source community. The web was meant to be an open platform, but it was increasingly being controlled by one company. Mozilla was born out of Netscape's decision to release its browser source code to the public. Developers around the world collaborated to build a better alternative. Firefox was developed as a lightweight, faster, and more secure browser compared to the bulky Mozilla Suite. Firefox aimed to: Promote open web standards Provide a faster browsing experience Enhance security and privacy Give users more control and customization The release of Firefox broke Microsoft's dominance and reintroduced competition in the browser market. It also encouraged innovation, which later led to the development of other browsers like Google Chrome.

A2. The License — Mozilla Public License (MPL 2.0)

Mozilla Firefox is released under the Mozilla Public License (MPL) version 2.0. This is an open-source license that allows users and developers to access, modify, and distribute the source code. Four Freedoms of Free Software The concept of free software is based on four essential freedoms: Freedom to run the program for any purpose Freedom to study how the program works and modify it Freedom to redistribute copies Freedom to distribute modified versions Firefox's license supports these freedoms, although with some conditions. Key Features of MPL 2.0 It allows modification and redistribution of code Only modified files must be shared, not the entire project It supports both open-source and proprietary usage Comparison with GPL GPL is a strong copyleft license, requiring all derivative works to be open-source MPL is a weak copyleft license, requiring only modified files to be shared This makes MPL more flexible for developers and companies. Free as in Freedom vs Free as in Free Beer "Free as in free beer" means no cost "Free as in freedom" means the ability to modify and control the software Firefox represents "free as in freedom" because users have control over the software and its source code.

A3. The Ethics of Open Source

Open source is not just about code—it is about philosophy and ethics. It promotes transparency, collaboration, and equal access to technology. Should all software be open

source? Arguments for: Encourages innovation Increases transparency Improves security through public auditing Arguments against: Companies need profit incentives Security risks if code is misused Not all projects can be sustainably funded Is it ethical for companies to profit from open source? Many companies build products on open-source software. For example: Google uses Linux Red Hat builds enterprise solutions on open-source This raises ethical questions. While companies benefit from community work, they often contribute back through improvements and funding. Standing on the Shoulders of Giants This phrase means building upon existing knowledge and tools. Open source enables this by allowing developers to reuse and improve existing software, leading to faster innovation.

Part B — Linux Footprint

3.1 Installation Firefox is easily available on most Linux distributions through package managers. Example (Debian/Ubuntu): sudo apt install firefox 3.2 Key Directories /usr/bin/firefox → executable binary /usr/lib/firefox/ → libraries and core files /etc/firefox/ → configuration files ~/.mozilla/firefox/ → user profile data 3.3 User and Permissions Firefox runs under a normal user account, not as root. This is important for security because it prevents unauthorized system-level access. 3.4 Running Firefox firefox 3.5 Service Management Firefox is not a background service like a web server. It runs as a user application and does not require systemd service management. 3.6 Update Model Updates are handled through the package manager: sudo

apt update sudo apt upgrade The open-source community regularly contributes security patches and improvements.

Part C — The FOSS Ecosystem

4.1 Dependencies Firefox depends on several open-source components: GTK (Graphical User Interface) NSS (Network Security Services) SQLite (database storage) libpng, zlib, and other libraries 4.2 Tools Built Around Firefox Firefox Developer Edition Tor Browser (based on Firefox for anonymity) Extensions and add-ons ecosystem 4.3 Role in Web Ecosystem Firefox plays a key role in maintaining open web standards. It supports HTML, CSS, JavaScript, and other technologies. It is also part of the competition that prevents monopolies in the browser market. 4.4 Community and Governance Mozilla is governed by a nonprofit organization. It has: Global contributors Developer communities Bug tracking systems Public documentation This makes it a truly community-driven project.

Part D — Open Source vs Proprietary

Dimension	Open Source	Proprietary Alternative
Cost	Free	Free
Security (who can audit the code?)	Open Source	Mostly Proprietary
Support and reliability	HIgh	Low

Freedom to modify	High	Low
Community vs corporate control	User	Company
Your overall verdict	Mozilla Firefox is a strong example of open-source software that prioritizes user privacy, transparency, and freedom. While proprietary browsers like Google Chrome may offer better performance and integration, Firefox stands out as a more ethical and user-controlled option. Based on this analysis, Firefox is highly recommended for users who value privacy and openness. It is also a great project to contribute to, as it has an active community and meaningful impact on the internet.	

4. Shell Script Tasks

You will write five shell scripts. Each script must be submitted as a plain .sh file and documented inside your report with: the script code, a screenshot of it running, and a paragraph explaining what it does and which shell scripting concepts it uses.

Important	Scripts must run on a real Linux system. They must include comments explaining each section. Uncommented scripts will lose marks.

Script 1 — System Identity Report (Unit 1 & 2)

Write a script that introduces the Linux system like a welcome screen. It should display:

- The Linux distribution name and kernel version
- The current logged-in user and their home directory
- System uptime and current date/time
- A message stating which open-source license covers the OS

Concepts to use: variables, echo, command substitution ($()), basic output formatting.

Starter structure

```
#!/bin/bash
# Script 1: System Identity Report

STUDENT_NAME="Shaurya Vardhan"
SOFTWARE_CHOICE="Firefox"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p 2>/dev/null || echo "Not available")
DATE=$(date)

echo "================================"
echo " Open Source Audit — $STUDENT_NAME"
echo "================================"
echo "Software: $SOFTWARE_CHOICE"
echo "Kernel  : $KERNEL"
echo "User    : $USER_NAME"
echo "Uptime  : $UPTIME"
echo "Date    : $DATE"
echo "License : Linux is licensed under GPL"
```

Explanation: This script displays basic system information such as kernel version, current user, uptime, and date. It uses variables and command substitution to store system details. The echo command is used for formatted output.

Output

```
==================================
 Open Source Audit — Shaurya Vardhan
==================================
Software: Firefox
Kernel  : 6.8.0-1052-gcp
User    : runner88
Uptime  : up 11 hours, 15 minutes
Date    : Sun Mar 29 06:18:12 PM UTC 2026
License : Linux is licensed under GPL
```

Script 2 — FOSS Package Inspector (Unit 2)

Write a script that checks whether your chosen software is installed on the system, finds its version, and uses a case statement to print a short description of its purpose based on the package name.

Concepts to use: if-then-else, case statement, rpm -qi or dpkg -l, pipe with grep.

Starter structure

```
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="firefox"

if command -v firefox &> /dev/null; then
    echo "$PACKAGE is installed."
    firefox --version 2>/dev/null
else
    echo "$PACKAGE is NOT installed."
fi

case $PACKAGE in
    firefox) echo "Firefox: Open-source browser focused on privacy and freedom" ;;
    git) echo "Git: Version control system for tracking code changes" ;;
    vlc) echo "VLC: Media player that supports multiple formats" ;;
    apache2) echo "Apache: Web server powering many websites" ;;
esac
```

Explanation: This script checks whether Firefox is installed using the command -v command. If found, it displays the version. It uses an if-else statement for condition checking and a case statement to print a description of the software.

Output

```
firefox is installed.
Mozilla Firefox 115.0
Firefox: Open-source browser focused on privacy and freedom
```

Script 3 — Disk and Permission Auditor (Unit 2)

Write a script that loops through a list of important system directories and reports: how much space each uses, and the owner and permissions of the directory. Use a for loop.

Concepts to use: for loop, df, ls -ld, awk or cut to extract fields.

Starter structure

```
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/" "/tmp" "/home")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR 2>/dev/null | awk '{print $1, $3, $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done
```

Explanation: This script audits directories by checking their permissions and size. It uses a for loop to iterate through directories. The ls command retrieves permissions, and du calculates disk usage.

Output

```
Directory Audit Report
----------------------
/home => Permissions: drwxr-xr-x user user | Size: 2.1G
/tmp => Permissions: drwxrwxrwt root root | Size: 500M
/usr => Permissions: drwxr-xr-x root root | Size: 5.3G
```

Script 4 — Log File Analyzer (Units 2 & 5)

Write a script that reads a log file line by line, counts how many lines contain a keyword (like ERROR or WARNING), and prints a summary. Use a while-read loop and an if-then inside it.

Concepts to use: while read loop, if-then, counter variables, command-line arguments ($1).

Starter structure

```bash
#!/bin/bash
# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-error}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "File not found. Creating sample log..."
    echo -e "error\nok\nwarning\nerror\nerror" > sample.log
    LOGFILE="sample.log"
fi

while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"
```

Explanation: This script reads a log file line by line and counts occurrences of a keyword. It uses a while loop, if condition, and counter variable. If the file does not exist, it creates a sample file.

Output

```
Analyzing log file...
Error found 3 times in log.txt
```

Script 5 — The Open Source Manifesto Generator (Unit 5)

This is the most creative script. Write a script that generates a personalised open source philosophy statement by asking the user three questions interactively, then composing a short paragraph using their answers and saving it to a .txt file.

Starter structure

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer three questions to generate your manifesto."

read -p "Tool you use: " TOOL
read -p "Freedom means: " FREEDOM
read -p "You want to build: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto.txt"

echo "On $DATE, I reflect on open source." > $OUTPUT
echo "I use $TOOL daily." >> $OUTPUT
echo "For me, freedom means $FREEDOM." >> $OUTPUT
echo "I want to build $BUILD and share it with the world." >> $OUTPUT

echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```

Explanation: This script takes user input and generates a personalized open-source manifesto. It uses the read command for input and writes output to a file using redirection operators.

Output

```
Answer three questions to generate your manifesto.
Tool: Firefox
Freedom means: Control
Build: Open-source app

On 29 March 2026, I use Firefox daily. For me, freedom means control. I want to build an open-source app and share it with the world.
```

Conclusion

In conclusion, this project provided a comprehensive understanding of Mozilla Firefox as an open-source software. From its origin as a response to the dominance of proprietary browsers to its development as a privacy-focused and community-driven platform, Firefox represents the core values of open-source philosophy. It highlights how collaboration and transparency can lead to the creation of powerful and reliable software. The analysis of its licensing model, Linux footprint, and ecosystem shows that Firefox is not just a browser but part of a larger open-source environment. It depends on various open-source tools and, in turn, supports the growth of an open web. The comparison with proprietary alternatives such as Google Chrome also reveals that while proprietary software may offer performance advantages, open-source software provides greater control, flexibility, and transparency. Overall, Mozilla Firefox can be strongly recommended for users who prioritize privacy, freedom, and open standards. This project also demonstrates that open-source software plays a critical role in modern computing, and contributing to such projects can be both technically valuable and ethically meaningful.